Ng, Kam Woh

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Research Interests

I am a Research Scientist at Meta AI London. My long-term goal is to make 3D worlds as easy to build as they are to imagine — so anyone, not only engineers, can freely express the worlds in their head. My research focuses on two intertwined directions:

Controllable Generative World Models: building models that learn to render, simulate, and explore visual worlds directly from data — and making them directable by users, rather than merely impressive.

Efficient Generative Architectures: designing new model architectures that make video generation real-time out of the box — without relying on complex post-training or distillation.

Experience

Meta AI — *London, England*
AI Research Scientist — *May 2025 – Present*

- Core contributor on **video diffusion** and **generative neural rendering** efforts. Lead and contribute to flagship research projects spanning architecture design, post-training, and controllable generation.
- Lead **real-time talking-avatar generation** (data, architecture, training).
- Lead **large-scale data processing pipeline for reference-to-video generation** (data, training).
- Lead **camera-controlled** and **template-controlled video generation**.
- **Kaleido** (ICLR 2026): sequence-to-sequence generative rendering model that synthesises novel views without explicit 3D or per-scene optimisation — a step toward a general-purpose visual world model.
- **VecGlypher** (CVPR 2026): fine-tunes multimodal LLMs to emit SVG path commands directly, producing editable vector graphics from text or image exemplars — evidence that LLMs already reason about 2D geometry.
- **Rays as Pixels** (ICML 2026): unifies video generation and camera-pose estimation by encoding cameras as 3-channel "raxels" so both modalities are co-denoised through a single video VAE.

Meta AI — *London, England*
Research Scientist Intern — *Aug 2024 – Apr 2025*

- Advisor: Dr. Xiao (Brandon) Han.
- Started the Kaleido line of work on generative neural rendering.
- Built large-scale Blender rendering and 3D-asset pipelines used to train and evaluate generative rendering models.
- Contributed to human pose transfer (for virtual try-on) and talking-avatar generation.

Huawei Noah's Ark Lab — *London, England*
Research Scientist Intern — *Feb 2024 – Jul 2024*

- Advisors: Dr. Jiankang Deng (Imperial College London) and Dr. Jing Yang (University of Cambridge).
- **Chirpy3D**: part-aware 3D generation from *unposed* 2D images, by fine-tuning multi-view diffusion with LoRA and distilling part-controlled views into NeRF.

WeBank — AI Department — *Shenzhen, China*
AI Research Engineer — *Nov 2019 – Mar 2021*

- Research focus on AI security in federated learning (privacy protection, adversarial defence, ownership verification).
- Lead author on **DPN** (IJCAI 2020) and **DeepIPR** (TPAMI 2022).
- Co-lead AAAI 2021 workshop *Towards Robust, Secure and Efficient Machine Learning*.
- Co-edited the Springer book *Federated Learning: Privacy and Incentive*.
- Co-author on **IPR-GAN** (CVPR 2021) and **IPR-Caption** (Pattern Recognition 2022) — dual

white-box / black-box watermarking for generative models and captioners.

Centre of Image and Signal Processing, University of Malaya *Kuala Lumpur, Malaysia*
Research Assistant *Feb 2019 – Oct 2019*

- Advisors: Prof. Chan Chee Seng (UM) and Dr. Lixin Fan (WeBank).
- **DeepIPR** (NeurIPS 2019, first publication): passport-based DNN ownership verification that defeats ambiguity attacks without degrading model performance.

CISIP Lab, University of Malaya *Kuala Lumpur, Malaysia*
Research Intern *Jul 2016 – Aug 2016*

- Built a deep-learning Malaysian car-plate recognition system (Caffe).

Education

University of Surrey — Centre for Vision, Speech and Signal Processing *Guildford, England*
Ph.D. in Computer Science *Oct 2021 – Jun 2025*

- Advisors: Prof. Tao Xiang and Prof. Yi-Zhe Song.
- Close collaborator: Dr. Xiatian (Eddy) Zhu.
- Thesis: Discrete Representation Learning for Retrieval and Generation — from interpretable hashing (ConceptHash) and orthogonal codebooks (OrthoHash, SDC) to compositional and part-aware generation (PartCraft, Chirpy3D).

University of Malaya *Kuala Lumpur, Malaysia*
B.Sc. in Computer Science (Artificial Intelligence), First Class *Aug 2015 – Jan 2019*

- CGPA: 3.89 / 4.00. Final year project (CNN interpretability) accepted at AAAI-19 workshop.

Selected Publications

*(**Bold** indicates Kam Woh Ng as an author. Full list on Google Scholar.)*

1. *Scaling Sequence-to-Sequence Generative Neural Rendering (Kaleido).*
   Shikun Liu, **Kam Woh Ng**, Wonbong Jang, Jiadong Guo, Junlin Han, Haozhe Liu, Yiannis Douratsos, Juan C. Pérez, Zijian Zhou, Chi Phung, Tao Xiang, Juan-Manuel Pérez-Rúa.
   International Conference on Learning Representations (ICLR), 2026.

2. *Rays as Pixels: Learning a Joint Distribution of Videos and Camera Trajectories.*
   Wonbong Jang, Shikun Liu, Soubhik Sanyal, Juan C. Pérez, **Kam Woh Ng**, Sanskar Agrawal, Juan-Manuel Pérez-Rúa, Yiannis Douratsos, Tao Xiang.
   International Conference on Machine Learning (ICML), 2026.

3. *VecGlypher: Unified Vector Glyph Generation with Language Models.*
   Xiaoke Huang, Bhavul Gauri, **Kam Woh Ng**, Tony Ng, Mengmeng Xu, Zhiheng Liu, Weiming Ren, Zhaochong An, Zijian Zhou, Haonan Qiu, Yuyin Zhou, Sen He, Ziheng Wang, Tao Xiang, Xiao Han.
   IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.

4. *Chirpy3D: Continuous Part Latents for Creative 3D Bird Generation.*
   Kam Woh Ng, Jing Yang, Jia Wei Sii, Jiankang Deng, Chee Seng Chan, Yi-Zhe Song, Tao Xiang, Xiatian Zhu.
   Preprint, 2024.

5. *PartCraft: Crafting Creative Objects by Parts.*
   Kam Woh Ng, Xiatian Zhu, Yi-Zhe Song, Tao Xiang.
   European Conference on Computer Vision (ECCV), 2024.

6. *ConceptHash: Interpretable Fine-Grained Hashing via Concept Discovery.*
   Kam Woh Ng, Xiatian Zhu, Jiun Tian Hoe, Chee Seng Chan, Tianyu Zhang, Yi-Zhe Song, Tao Xiang.
   CVPR Workshop on Fine-Grained Visual Categorization (FGVC11), 2024. **[Best Paper Award]**

7. *Unsupervised Hashing with Similarity Distribution Calibration (SDC).*
   Kam Woh Ng, Xiatian Zhu, Jiun Tian Hoe, Chee Seng Chan, Tianyu Zhang, Yi-Zhe Song, Tao Xiang.

British Machine Vision Conference (BMVC), 2023. **[Oral]**

8. *One Loss for All: Deep Hashing with a Single Cosine Similarity-Based Learning Objective (OrthoHash).*
   Jiun Tian Hoe, **Kam Woh Ng**, Tianyu Zhang, Chee Seng Chan, Yi-Zhe Song, Tao Xiang.
   Neural Information Processing Systems (NeurIPS), 2021.
9. *Deep Polarized Network for Supervised Learning of Accurate Binary Hashing Codes (DPN).*
   Lixin Fan, **Kam Woh Ng**, Ce Ju, Tianyu Zhang, Chee Seng Chan.
   International Joint Conference on Artificial Intelligence (IJCAI), 2020.
10. *Rethinking Deep Neural Network Ownership Verification: Embedding Passports to Defeat Ambiguity Attacks (DeepIPR).*
    Lixin Fan, **Kam Woh Ng**, Chee Seng Chan.
    NeurIPS, 2019; extended journal version in IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2022.

Honors, Awards & Competitions

Research Honors & Awards

- *2024* **Best Paper Award** CVPR Workshop on Fine-Grained Visual Categorization (FGVC11).
- *2023* **Oral Presentation** British Machine Vision Conference (BMVC).
- *2019* **Dean's Special Award** Faculty of Computer Science & IT, University of Malaya.
- *2019* **Excellent Project Award** Faculty of Computer Science & IT, University of Malaya.

Competitions & Hackathons

- *2024* **Second Place** FishNet Competition, ECCV 2024.
- *2022* **Second Place** eBay eProduct Visual Search Challenge, FGVC9 (CVPR 2022).
- *2019* **36th Place** IKCEST Big Data Challenge.
- *2018* **Top 10** Grab AI for S.E.A. Challenge.
- *2018* **First Prize** E-Genting Programming Competition.
- *2018* **Second Prize** ACM-ICPC Malaysia al-Khawarizmi National Programming Contest.
- *2018* **Third Prize** Prosolve National Programming Competition.
- *2018* **4th Place** KPMG Security Challenge Malaysia.
- *2018* **Second Prize** HEREMYHACK Virtual Hackathon.
- *2018* **Open Cat. Winner** Park of the Future Hackathon.
- *2017* **Second Runner-Up** UNICODE Programming Contest.
- *2017* **First Runner-Up** ATURKREATIF'17 Open Programming Competition.
- *2017* **Second Prize** E-Genting Programming Competition.
- *2017* **Top 5** F-Secure Intervarsity Cyber Security Competition.
- *2017* **First Prize** Sunway CityHack.
- *2016* **First Prize** Unlock Asia AI Robot & Big Data Hackathon.

Mentorship & Advising

Meta AI — Research Scientist Intern Co-Host *London, England*
Hosting visiting Ph.D. research scientist interns *Summer 2026*

- **Runjia Li** (Ph.D. at Oxford; co-hosted with Shikun Liu).
  Multi-player video world modelling.
- **Xiaoke Huang** (Ph.D. at UC Santa Cruz; co-hosted with Xiao (Brandon) Han).
  Training language models for vector glyph generation (**VecGlypher**, CVPR 2026).

Teaching

University of Surrey — Centre for Vision, Speech and Signal Processing
Teaching Assistant *Feb 2023 – Dec 2023*

- EEEM071 Advanced Topics in Computer Vision and Deep Learning.

– EEEM066 Fundamentals of Machine Learning.
– EEEM077 AR, VR and the Metaverse.

University of Malaya — Faculty of Computer Science and Information Technology
Teaching Assistant *Sep 2016 – Dec 2018*

– WIX1002 Fundamentals of Programming.
– WIA1002 Data Structures.

Service

Conference Reviewer

– *2026* CVPR, ECCV, SIGGRAPH, 3DV.
– *2025* CVPR, NeurIPS, ICLR, ICML, SIGGRAPH Asia, ACM MM, BMVC.
– *2024* ICML, AAAI.
– *2023* CVPR, ICCV, NeurIPS, AAAI, BMVC.
– *2022* CVPR, ECCV, ICME.
– *2021* CVPR, AAAI.

Journal Reviewer

– *2022* IEEE Signal Processing Letters.

Workshop & Book Organising

– *2021* **Organising Committee** AAAI Workshop on *Towards Robust, Secure and Efficient Machine Learning*.
– *2020* **Organising Committee** Springer book *Federated Learning: Privacy and Incentive*.

Technical Proficiency

Programming
Python, C/C++, C#, Java, JavaScript, GDScript.*

Frameworks
PyTorch, TensorFlow, Blender*, Unity3D, Godot, ReactJS.*

Languages
English, Mandarin, Cantonese, Malay.
(*) sufficient level of expertise.